

Below is a guideline for the grading rubric that the advisor (and co-advisor) will develop for the research report. While there is some flexibility in your adaptation of this rubric, the main categories cannot be removed. The **overall structure of the report**, the length requirements, the main categories within the Introduction, Methods, Results, Discussion section **should be preserved**. However, making the descriptions more specific to the project is encouraged. This could include number of references, expectations around reproducibility, expectations around results (e.g., type, quantity, and quality of figures), expectations about how the introduction and discussion should be structured, whether some of the specifications should move up or down a level, ...

In addition, if you want to use a more traditional grading rubric grid because you are more familiar with that, please do, but incorporate the components outlined below in it. If you have questions or concerns, please contact the course coordinator.

Bioinformatics Report: Bundle Specifications

Bundle C (70-80): The Foundational Level

Minimum requirements to demonstrate basic competency.

- **Abstract is not necessary!**
- **Introduction:** Provides sufficient set-up and literature review that leads to a knowledge gap and the biological question/hypothesis, how bioinformatics tools will solve this, and provides a basic explanation of its importance and novelty.
- **Methods:**
 - Identifies the final datasets used, including source, size, and key characteristics.
 - Lists the primary programming language and appropriate libraries/packages/pipelines.
 - Provides a high-level description of the analyses (bioinformatics, statistics, biological, ...).
- **Results:** Includes at least two primary deliverables (e.g., a data visualization and a summary table) that directly relate to the biological question/hypothesis.
- **Discussion:**
 - Summarizes the main findings of the research.
 - Discusses their importance in the field of research
- **Formatting:** The report is approximately 15 pages (excluding references/appendices, figures and tables), is professionally formatted, and includes a complete list of references, reference style determined by advisor

Bundle B (80-90): The Practitioner Level

Includes all requirements from Bundle C, plus the following:

- **Hypothesis Integration:** Evaluates and integrates multiple hypotheses/objectives/pipelines/...

- **Technical Rigor:**
 - Includes a detailed "Materials and Methods" section describing data cleaning, normalization, and any pre- and post-processing steps taken
 - Specifies and justifies the statistical tests or algorithms used to validate the results
 - Figures and tables can stand on their own, with clear legends, labels, and visual elements
- **Discussion:** Provides a comprehensive interpretation of the results, both from a bioinformatics and biology perspective, explaining the "so what?" factor.

Bundle A (90+): The Researcher Level

Includes all requirements from Bundles C and B, plus the following:

- **Novelty and complexity:** Of the biological question, bioinformatics approach, pipeline development, statistical analyses, narrative of the report, quality of the data created, quality of the figures and tables, ...
- **Reproducibility:** Includes a section on computational reproducibility, specifying how environments are managed (e.g., Conda, Docker, or a specific GitHub repository structure).
- **Advanced Analysis:**
 - Weaknesses and scope of sampling procedure and/or data collection procedures are described
 - Assumptions behind methodologies are described and defended
- **Critical Evaluation:** Provides a robust discussion of potential pitfalls, technical limitations, how biological noise might impact the findings, biological implications are discussed in depth, realistic ideas for future studies, ...
- **Professionalism:** Writing is of "submission quality," requiring zero to minimal editing, with perfectly formatted citations and high-resolution conceptual diagrams (if any).

The "Unsatisfactory / Fail" Category

A grade of **F** (or Unsatisfactory) is assigned if the proposal meets any of the following conditions:

- **Incomplete:** Fails to meet the basic requirements of **Bundle C**.
- **Plagiarism:** Contains unoriginal work or fails to cite data sources and literature.
- **Formatting breach:** Significantly misses the 15-page length requirement (either too short to cover the material or excessively long)